



Sample Set (Gallop Rules)



- This round contains 27 problems to be solved in 60 minutes.
- Problems are divided into 9 sets of 3 problems each.
- You must submit your current set before moving onto the next one.
- Make sure to pencil in (or pen in) all answers correctly on the answer sheet as you will be unable to return to a set once you have turned it in!
- Point values for each set:

Round #	1	2	3	4	5	6	7	8	9	Total Pts.
Pts/Problem	10	11	12	13	14	16	18	21	25	420

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Gallop Colt Round Set 1 Answer Sheet

Team ID _____ Team Name _____

Room # _____ Student Name(s) _____

1.

2.

3.

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Gallop Colt Round Set 1

1. [10] What is $20 + 25 + 20 \cdot 25 + 20 - 24 + 2025$?
2. [10] Right trapezoid $ABCD$ has a height of 6, and the parallel sides have lengths 6 and 14. What is the length of the last side?
3. [10] Consider an integer N that is formed by the first 100 positive integers in a row (as shown below). How many times in N does the string 19 appear?

$$N = 123456789101112\dots$$

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Gallop Colt Round Set 2 Answer Sheet

Team ID _____ Team Name _____

Room # _____ Student Name(s) _____

4.

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Gallop Colt Round Set 2 Answer Sheet

Team ID _____ Team Name _____

Room # _____ Student Name(s) _____

4.

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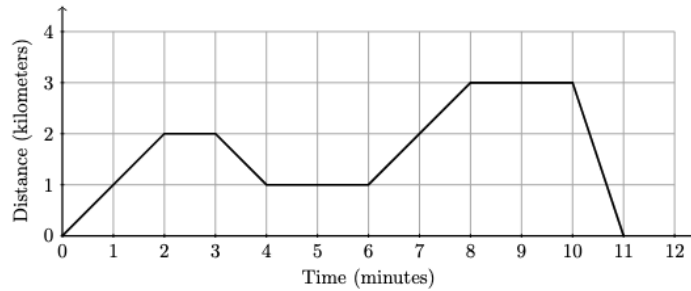
6.

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Gallop Colt Round Set 2

4. [11] Inside of regular hexagon $ABCDEF$, the area of triangle BDF is 25. What is the area of hexagon $ABCDEF$?
5. [11] Danny is walking on a straight path. The graph below shows his distance, in kilometers, from his starting point. For how many minutes was Danny actively walking?



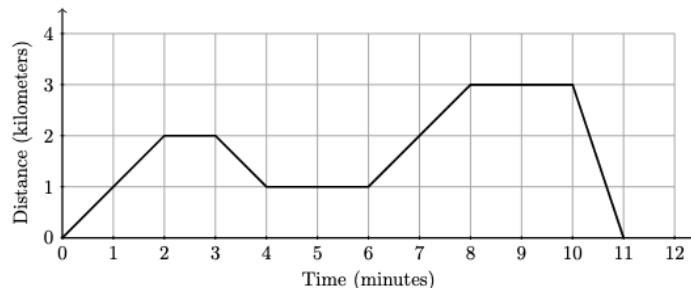
6. [11] Bryan loves Sparkling Ice Kiwi Strawberry flavored drinks. Each bottle costs \$1.99 pretax with a 10% sales tax. If Bryan has \$25, how many bottles can he buy?

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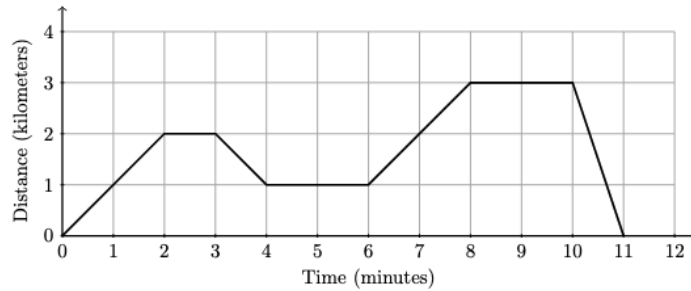


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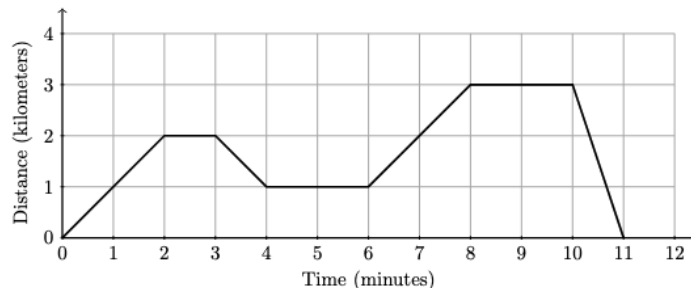
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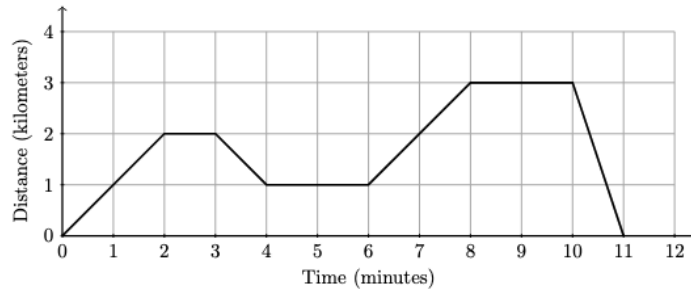


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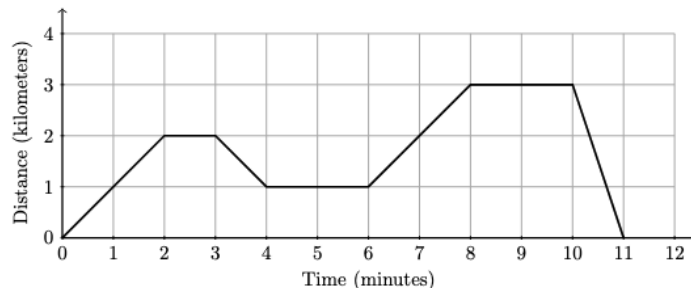
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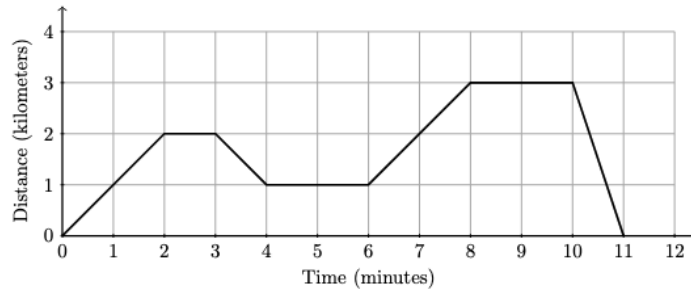


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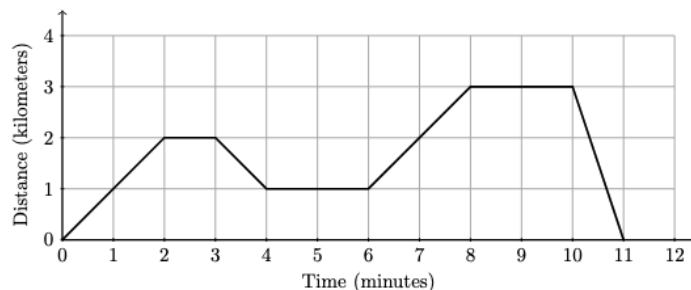
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7.

8.

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7.

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9.

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Gallop Colt Round Set 3

7. [12] How many ways are there to split 100 indistinguishable fossils between Alice and Bob such that each gets at least one fossil?
8. [12] How many integers less than 500 can be written in the form $2^a 3^b$, where $a \geq 0$ and $b \geq 0$?
9. [12] What is the remainder when $2^{2025} + 0^{2025} + 2^{2025} + 5^{2025}$ is divided by 25?

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10.

11.

12.

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Gallop Colt Round Set 4 Answer Sheet

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10.

11.

12.

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Gallop Colt Round Set 4

10. [13] Benjamin has gotten the following scores on the 25 question MMMA10 (Mustang Math Mock AMC10): 67.5, 90, 100.5, 106.5. Mustang Math grants an honor of “Extremely and Highly Distinguished Honor Roll” to people who average over a score of 100 over the course of at least 5 years. What is the minimum number of questions correct Benjamin can get on his next MMMA10 if he wants to get “Extremely and Highly Distinguished Honor Roll?” Note that correct answers are worth 6 points, incorrect 0 points, and blank 1.5 points.
11. [13] Alan likes to listen to music, so he brings his wireless earbuds to school every day. However, the earbuds are very small and easy to misplace, so Alan has a 25% chance of losing his earbuds every day at school. What is the probability that Alan still has his earbuds by the end of a school week (5 days)? Give your answer as a simplified fraction.
12. [13] What is the sum of the solutions to $\sqrt{x^2 + 6x + 9} = 3.1415$?

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Gallop Colt Round Set 5 Answer Sheet

Team ID _____ Team Name _____

Room # _____ Student Name(s) _____

13.

14.

15.

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Gallop Colt Round Set 5 Answer Sheet

Team ID _____ Team Name _____

Room # _____ Student Name(s) _____

13.

14.

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Gallop Colt Round Set 5

13. [14] If $f(2x) = \frac{2}{2+x}$ for all $x > 0$, and if $2f(x) = \frac{a+bx}{c+dx}$, such that a, b, c, d are all non-negative integers, what is the minimum value of $a + b + c + d$?
14. [14] What is the greatest integer less than $\sqrt{28} + \sqrt{45}$?
15. [14] Billy started his Spanish learning streak on Duolingo on November 24, 2024, a Sunday. If he studies every day, when will he reach his 100 day streak? Give your answer in the form of MM/DD, including leading zeros if necessary.

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Gallop Colt Round Set 6 Answer Sheet

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Gallop Colt Round Set 6 Answer Sheet

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Gallop Colt Round Set 6

16. [16] Find the number of positive integers n less than 200 such that the greatest common divisor of n and 60 is 6.
17. [16] Pablo Panther and Roger Rabbit are playing a game with coins. Pablo starts with n coins, where $1 < n < 1000$. On Pablo's turn, if Pablo has an odd number of coins, his coins are tripled, and Roger gives him 1 extra coin on top of his earnings. However, if Pablo has an even number of coins, he loses half of his coins. Let $f(x)$ represent the number of coins Pablo has after x moves. Find the value of $f(15411) + f(15412) + f(15413)$.
18. [16] If a , b , and c are positive digits with a sum ≤ 10 , what is the positive four-digit integer $\overline{a0bc}$ that is equal to the product of the sum of its digits and $\overline{abc}$?

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Gallop Colt Round Set 7 Answer Sheet

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19.

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Gallop Colt Round Set 7

19. [18] X and Y are two points that are 6 units apart. Point Z lies on line $\overline{XY}$ such that $\overline{XZ} = 2\overline{YZ}$. What is the sum of all the possible values of $\overline{XZ}$?
20. [18] M is the set of divisors of 2025, while T is the set of integers from 1 to 2025. What is the probability that a randomly selected element from M and a randomly selected element from T are not coprime?
21. [18] In how many ways can a committee of 5 members be formed from 6 men and 8 women such that the committee contains at least 3 women?

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Gallop Colt Round Set 8

22. [21] What is the sum of all integers x such that $(x^2 + 9x + 19)^{2x^2 + 8x + 8} = 1$?
23. [21] Triangle ABC has $AB = 13$ and $BC = 15$. Point D lies on side AC such that $AD = 10$ and $BD = 13$. What is CD ?
24. [21] A fully-charged flashlight that is continuously on lasts for 12 hours. In the morning, the flashlight has a full battery and is shined until the battery reaches 25 percent. It is then turned off until the battery dies. This process took a total of 19 hours. How many hours does it take for a turned-off flashlight with a full battery to drain to 50 percent? Assume the rate of battery drain is constant.

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Gallop Colt Round Set 9 Answer Sheet

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Gallop Colt Round Set 9 Answer Sheet

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Gallop Colt Round Set 9

25. [25] Three coins are placed heads-up on a table. Each minute, Bob randomly selects one coin and flips it (heads to tails or tails to heads). What is the expected number of heads remaining after five minutes?
26. [25] How many functions $\{1, 2, 3, 4\} \Rightarrow \{1, 2, 3, 4\}$ satisfy $2f(x) \geq x + f(f(x))$ for all x .
27. [25] On a circle with diameter $AB = 2r$, where r is an integer, chords parallel to AB have lengths $2r - 2$ and $2r - 4$. Find the sum of the two smallest values of r such that the distance between each pair of chords (including AB) is an integer.

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27. [25] On a circle with diameter $AB = 2r$, where r is an integer, chords parallel to AB have lengths $2r - 2$ and $2r - 4$. Find the sum of the two smallest values of r such that the distance between each pair of chords (including AB) is an integer.

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Gallop Colt Round Set 9

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